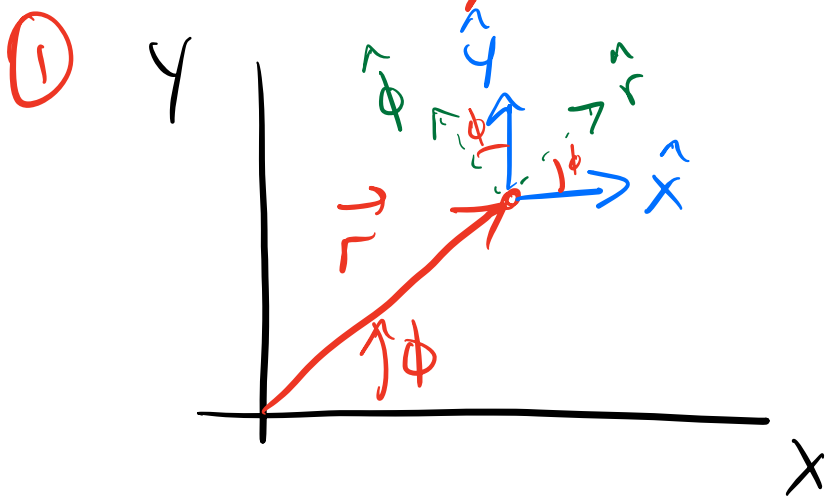


Deriving Newton's 2nd Law in Plane Polar

1. Draw $\vec{r}$
2. compute $d\vec{r}/dt = \vec{v}$
3. compute $d^2\vec{r}/dt^2 = \vec{a}$
4. investigate $\vec{F} = m\vec{a}$



$$\vec{r} = r \hat{r} \quad \text{where } r = |\vec{r}|$$

②

$$\frac{d\vec{r}}{dt} = \underbrace{\frac{dr}{dt}}_{\dot{r}} \hat{r} + r \underbrace{\frac{d\hat{r}}{dt}}_{\dot{\hat{r}}}$$

do in cartesian

$$\begin{aligned} \hat{r} &= \cos\phi \hat{x} + \sin\phi \hat{y} \\ \hat{\phi} &= -\sin\phi \hat{x} + \cos\phi \hat{y} \end{aligned}$$

$$\hookrightarrow \frac{d\hat{r}}{dt} = -\dot{\phi} \sin\phi \hat{x} + \dot{\phi} \cos\phi \hat{y} = \dot{\phi} \hat{\phi}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \dot{r} \hat{r} + r \dot{\phi} \hat{\phi}$$

$$\textcircled{3} \quad \vec{a} = \frac{d\vec{v}}{dt} = \underbrace{\frac{d\dot{r}}{dt}}_{\ddot{r}} \hat{r} + \underbrace{\dot{r} \frac{d\hat{r}}{dt}}_{\dot{\phi} \hat{\phi}} + \underbrace{\frac{dr}{dt} \dot{\phi} \hat{\phi}}_{\dot{r} \dot{\phi} \hat{\phi}} + \underbrace{r \frac{d\dot{\phi}}{dt} \hat{\phi}}_{\ddot{\phi} \hat{\phi}} + \underbrace{r \dot{\phi} \frac{d\hat{\phi}}{dt}}_{?}$$

$$\frac{d\hat{\phi}}{dt} = -\dot{\phi} \cos \phi \hat{x} - \dot{\phi} \sin \phi \hat{y} = -\dot{\phi} \hat{r}$$

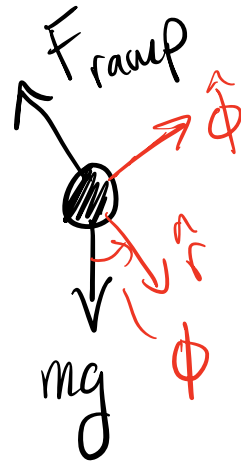
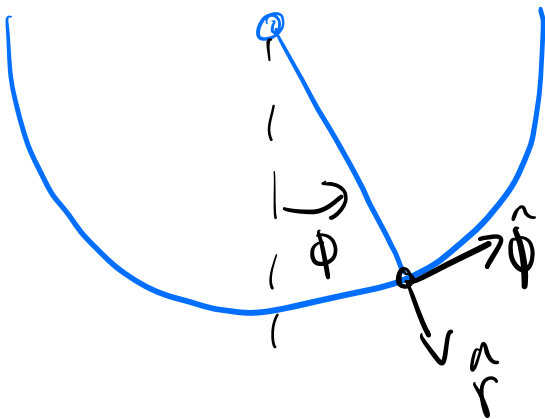
$$\vec{a} = \ddot{r} \hat{r} + \dot{r} \dot{\phi} \hat{\phi} + \dot{r} \dot{\phi} \hat{\phi} + r \ddot{\phi} \hat{\phi} + r \dot{\phi} (-\dot{\phi} \hat{r})$$

$$\vec{a} = (\ddot{r} - r \dot{\phi}^2) \hat{r} + (r \ddot{\phi} + 2\dot{r} \dot{\phi}) \hat{\phi}$$

$$\textcircled{4} \quad \vec{F}_{net} = m \vec{a} \Rightarrow \vec{F}_r + \vec{F}_\phi = m(\vec{a}_r + \vec{a}_\phi)$$

$$F_r = m(\ddot{r} - r \dot{\phi}^2) \quad F_\phi = m(r \ddot{\phi} + 2\dot{r} \dot{\phi})$$

Skateboard Example



$$\sum F_r = ma_r = -F_{\text{ramp}} + mg \cos \phi = m(\ddot{r} - r\dot{\phi}^2)$$

Note $r=R$ so $\ddot{r}=0$,

$$-F_{\text{ramp}} + mg \cos \phi = -mR\dot{\phi}^2$$

$$\sum F_{\phi} = ma_{\phi} = -mg \sin \phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi})$$

Note: $r=R$ $\dot{r}=0$

$$-mg \sin \phi = mR\ddot{\phi}$$

or

$$\ddot{\phi} = -\frac{g}{R} \sin \phi$$

Assume small osc.

$$\sin\phi \approx \phi$$

$$\hookrightarrow \ddot{\phi} = -\frac{g}{R}\phi \Rightarrow \ddot{x} = -\omega^2 x?$$

$$\omega^2 = g/R \quad \Rightarrow \quad \omega = \sqrt{g/R}$$

$$\phi(t) = A \cos(\omega t) + B \sin(\omega t) \quad \text{general soln.}$$

Lagrangian Dynamics

①

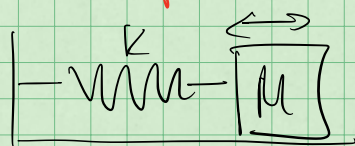
So far you have studied much of physics from the perspective of Newtonian Mechanics. The idea behind that is

- ① we can identify all the interactions with a body (i.e. the forces & torques)
- ② we can write models of those interactions using mathematical functions (i.e. $F = -kx$, βv^2)
- ③ We vectorially add all the individual interactions to find the net result (i.e. $\sum \vec{F}_i = \vec{F}_{net}$)
- ④ We apply Newton's Second to the body

$$m \vec{\ddot{x}} = \vec{F}_{net}$$

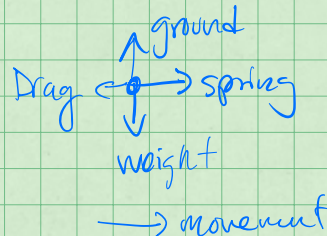
this is our equation of motion (EOM)

Quick Example: 1D SHO w/ linear drag



B drag coeff. ① Interactions?

- ② -ground = weight = $-mg$ ($\hat{y}$)
 spring = $-kx$ Drag = $-\beta \dot{x}$ ($\hat{x}$)



③ $\sum \vec{F}_i \rightarrow$ weight + ground = 0 nothing interesting
 $\rightarrow$ spring + drag = $F_{\text{net},x}$ ②

$$F_{\text{net},x} = -kx - \beta \dot{x}$$

④ $m\ddot{x} = -kx - \beta \dot{x} \Rightarrow \boxed{m\ddot{x} + \beta \dot{x} + kx = 0}$

What about Lagrange?

The Lagrange Formulation is rooted in classical optimization, but it is equivalent to Newton's work. However, it uses energy (a scalar) to do so. This means we can exploit coordinate transformations that can't change the scalar value (of energy).

Epistemologically speaking, Lagrange's approach is an exploration of phase space to determine paths the dynamical system can take. You can think of this as one level above plotting the known phase space, b/c we don't yet know the EOMs.

Let's do an example and then backup.

Lagrange SHO

(3)

As you will learn to get the EOM for a non-dissipative system we form a function called the Lagrangian,

$$L = T - U \quad \text{where} \quad \begin{array}{l} T = \text{kinetic energy} \\ U = \text{potential energy} \end{array}$$

$$T = +\frac{1}{2}m\dot{x}^2 \quad U = +\frac{1}{2}kx^2 \quad \text{1D SHO}$$

Now we follow the optimization routine,

$$\frac{dL}{dx} - \frac{d}{dt} \left(\frac{dL}{d\dot{x}} \right) = 0 \quad \text{1D Euler-Lagrange Equ}$$

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2 \quad \frac{dL}{dx} = -kx \quad \frac{dL}{d\dot{x}} = m\dot{x}$$

thus,

$$-kx - \frac{d}{dt}(m\dot{x}) = -kx - m\ddot{x} = 0$$

$$\text{so, } m\ddot{x} + kx = 0 \quad \text{or } \ddot{x} = -\frac{k}{m}x$$

MAGIC?

no just framing mechanics differently

Calculus of Variations

(4)

This is not magic; it's an application of an approach that concludes of letting a system evolve so long as it starts from given location in phase space and moves to another given location.

+ We find the "optimal" path.

⇒ This is called "variational calculus" or "calculus of variations"

* A discussion of the origins and base theory are not here. But references are included (Boas, Weber, Arfken, Goldstein all cover that).

Statement of the Calc of Variation Problem

Consider an integral J of the form,

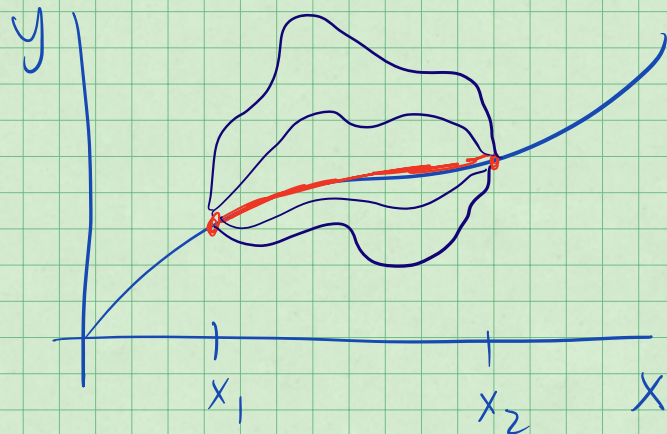
$$J = \int_{x_1}^{x_2} f(y, y') dx \quad \text{where } y = y(x) \\ y' = \frac{dy(x)}{dx}$$

We aim to determine how we can choose f so that J is an extremum (i.e. a min or max)
typically min in mechanics

Thought Experiment

(5)

Assume we know the path through phase space



true path
(an extremum
of J)

alternative paths
with larger J 's.

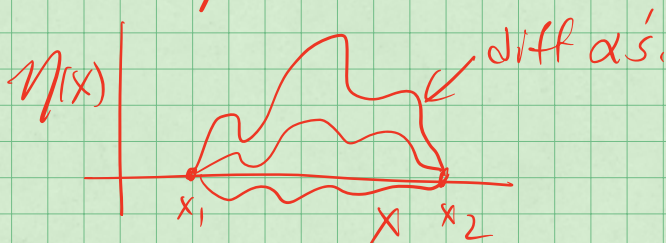
$$y(\alpha, x) = y(0, x)$$

where $\alpha=0$ represents
the known path

$$y(\alpha, x) = y(0, x) + \alpha \eta(x)$$

note: $\eta(x_1) = 0$
 $\eta(x_2) = 0$

where $\eta(x)$ is additional
function describing possible
alt. paths.



Finding the extrema means taking $\left. \frac{dJ}{d\alpha} \right|_{\alpha=0}$
and setting that to zero.

$$J(\alpha) = \int_{x_1}^{x_2} f(y(\alpha, x), y'(\alpha, x)) dx$$

I won't derive in detail. It's in most books that taking $dJ/dx|_{x=0} = 0$ (6)

gives

$$\left[\frac{df}{dy} - \frac{d}{dx} \left(\frac{df}{dy} \right) = 0 \quad \text{Euler equations} \right]$$

Example: Brachistochrone (in class)

Hamilton's Principle

The path followed by a dynamical system minimizes the time integral of $T-U$.

Mathematically,

$$\delta \int_{t_1}^{t_2} (T-U) dt = 0$$

with

$$T = T(\dot{x}_i) \quad U = U(x_i)$$

$$L \equiv T - U = L(x_i, \dot{x}_i)$$

So

$$\delta \int_{t_1}^{t_2} L(x_i, \dot{x}_i) dt = 0$$

such that

$$\left[\frac{dL}{dx_i} - \frac{d}{dt} \left(\frac{dL}{d\dot{x}_i} \right) = 0 \quad i=1,2,3, \right]$$

We have Lagrange's EoM! for cartesian x_i (7)

$$\frac{dL}{dx_i} - \frac{d}{dt} \left(\frac{dL}{d\dot{x}_i} \right) = 0 \quad i=1, 2, 3$$

$\boxed{x, y, z}$ think

Generalized Coordinates

As it turns out, Lagrangian Dynamics can make use of any phase coordinate ($\theta, \phi, x, y, \rho, z$, etc.) and its first derivative ($\dot{\theta}, \dot{\phi}, \dot{x}, \dot{y}, \dot{\rho}, \dot{z}$, etc.)

So In fact for any set of general coordinates,

$$\vec{q} = \langle q_1, q_2, q_3, \dots, q_N \rangle$$

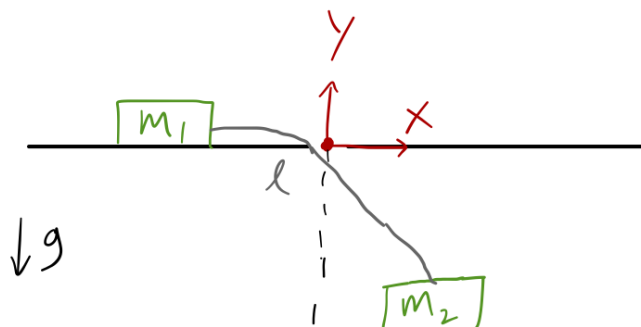
and their derivatives,

$$\dot{\vec{q}} = \langle \dot{q}_1, \dot{q}_2, \dot{q}_3, \dots, \dot{q}_N \rangle$$

give rise to a general set of N equations (subject to reductions due to constraint eqns. (more later))

$$\frac{dL}{dq_i} - \frac{d}{dt} \left(\frac{dL}{d\dot{q}_i} \right) = 0 \quad i=1, 2, \dots, N$$

Tutorial: Lagrangian Mechanics



Two masses m_1 and m_2 are connected by a massless rope of length ℓ . As pictured, one of them rests on a frictionless table, while the other is dangling over the edge. We'll treat this as a **two-dimensional** system, starting with Cartesian coordinates (x, y) as depicted. Note that in particular, this means that **block 2 is free to swing back and forth!**

I. Choosing coordinates

A. Using the Cartesian coordinates of the two blocks (x_1, y_1) and (x_2, y_2) , write down a set of equations which represent all of the constraints on the system as given.

B. How many degrees of freedom does this system have?

C. Sketch a set of generalized coordinates on the diagram above.

D. [Discussion] By treating this problem as two-dimensional, we are *ignoring* the coordinates z_1 and z_2 , knowing that if $z_1(0) = z_2(0) = 0$, the system doesn't move in the z -direction. What about the coordinate x_2 - is it *ignorable*? Why or why not?

II. Setting up the Lagrangian

On the board, we've set up a common set of generalized coordinates (GCs) to use, and re-expressed the Cartesian coordinates using them. Let's go on to write down the Lagrangian describing this system.

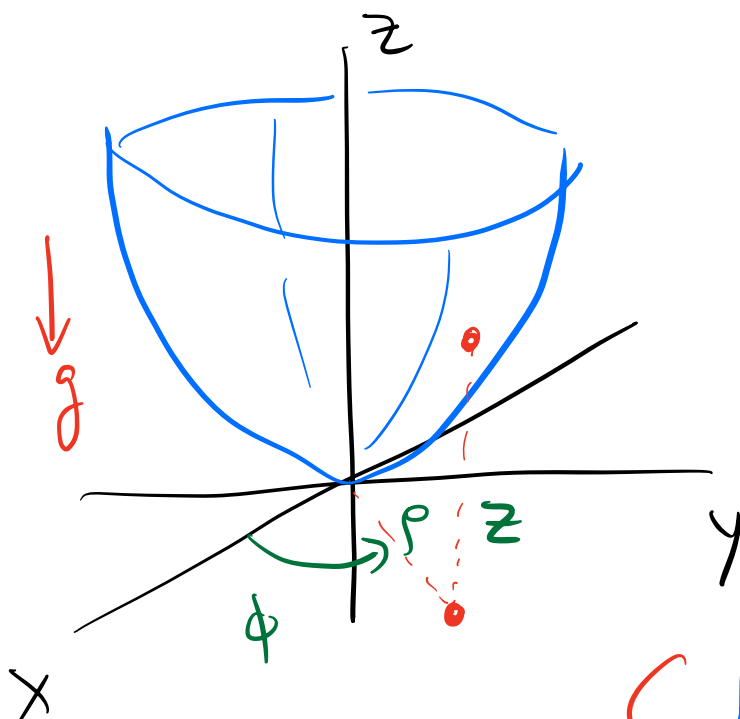
A. Write the potential energy U in terms of the generalized coordinates.

B. Evaluate the time derivatives of the Cartesian coordinates in terms of the GCs (and their derivatives), and use your results to write the kinetic energy T .

C. Combine your answers from A and B to obtain the Lagrangian, $\mathcal{L}$.

D. *[Discussion]* Find the equations of motion for both GCs by applying the Euler-Lagrange equations. What do you notice about the equations of motion for the two GCs? What does this tell you about the motion of the system?

Bead in a paraboloid



cylindrical coordinates

$$\langle \rho, \phi, z \rangle$$

$$v^2 = \dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2$$

$$z = c\rho^2$$

Important for
sensemaking

Note c has units

$$\begin{aligned} [z] &= m \\ [\rho^2] &= m^2 \quad \therefore [c] = \frac{1}{m} \end{aligned}$$

$$T = \frac{1}{2} m v^2 \Rightarrow \text{in cylindrical}$$

$$T = \frac{1}{2} m (\dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2) \quad \text{pt. particle kinetic energy}$$

$$U = mgz \quad \text{grav potential energy (pt. particle near Earth)}$$

$$\mathcal{L} = \frac{1}{2} m (\dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2) - mgz$$

$$\mathcal{L}(\rho, \dot{\rho}, \phi, \dot{\phi}, z, \dot{z}, t) \quad \begin{array}{l} \text{constraint} \\ z = c\rho^2 \\ \dot{z} = 2c\rho\dot{\rho} \end{array}$$

$\Downarrow$

$$\mathcal{L}(\rho, \dot{\rho}, \phi, \dot{\phi}, z) \quad \text{but really } \mathcal{L}(\rho, \dot{\rho}, \dot{\phi})$$

no ϕ dependence $\Rightarrow$ expect L conservation

no explicit t dependence $\Rightarrow$ expect Energy conservation

$$\mathcal{L} = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + (2c\rho\dot{\rho})^2) - mgc\rho^2$$

$$= \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + 4c^2\rho^2\dot{\rho}^2) - mgc\rho^2$$

EqM for ϕ why? b/c only $\dot{\phi}^2$ term

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) = 0$$

$$\underbrace{0}_0 \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) = \frac{d}{dt} \left(\underbrace{m\rho^2\dot{\phi}}_{L_z} \right) = 0$$

z component of ang-mom
is conserved!

L_z

Eqn for ρ $\mathcal{L} = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + 4c^2\rho^2\dot{\rho}^2) - mgc\rho^2$

$$\frac{d\mathcal{L}}{d\rho} - \frac{d}{dt} \left(\frac{d\mathcal{L}}{d\dot{\rho}} \right) = 0$$

$$\frac{d\mathcal{L}}{d\rho} = m\rho\dot{\phi}^2 + 4c^2\rho\dot{\rho}^2 - 2mgc\rho$$

$$\frac{d\mathcal{L}}{d\dot{\rho}} = m\dot{\rho} + 4mc^2\rho^2\dot{\rho}$$

$$\frac{d}{dt} \left(\frac{d\mathcal{L}}{d\dot{\rho}} \right) = m\ddot{\rho} + 4mc^2(2\rho\dot{\rho}^2 + \rho^2\ddot{\rho})$$

So,

$$\cancel{m\ddot{\rho}} + 4\cancel{mc^2}(2\rho\dot{\rho}^2 + \rho^2\ddot{\rho}) = \cancel{m\rho\dot{\phi}^2} + 4\cancel{c^2\rho\dot{\rho}^2} - 2\cancel{mgc\rho}$$

$$\ddot{\rho}(1 + 4c^2\rho^2) + 8c^2\rho\dot{\rho}^2 = \rho\dot{\phi}^2 + 4c^2\rho\dot{\rho}^2 - 2gc\rho$$

$$\ddot{\rho}(1 + 4c^2\rho^2) + 4c^2\rho\dot{\rho}^2 - \rho\dot{\phi}^2 + 2gc\rho = 0$$

also

$$\frac{d}{dt} (m\rho^2\dot{\phi}) = 0$$

$$\text{or } 2\rho\dot{\rho}\dot{\phi} + \rho^2\ddot{\phi} = 0$$

Clean up into $\ddot{\rho} = f(\rho, \dot{\rho}, \dot{\phi}, t)$

$$\ddot{\phi} = g(\dot{\phi}, t)$$

$$\ddot{\rho} = \frac{\rho \dot{\phi}^2 - 4c^2 \rho \dot{\rho}^2 - 2cgr}{(1 + 4c^2 \rho^2)}$$

$$\ddot{\phi} = -\frac{2\rho \dot{\rho} \dot{\phi}}{\rho^2} \Rightarrow \text{solve away from } \langle 0, 0, 0 \rangle$$

$$\ddot{\phi} = -\frac{2\dot{\rho} \dot{\phi}}{\rho}$$

Prepare for Numerical Integration

Let $\omega = \dot{\phi}$ and $v = \dot{\rho}$ then,

$$\dot{v} = \frac{\rho \omega^2 - 4c^2 \rho v^2 - 2cgr}{(1 + 4c^2 \rho^2)}$$

$$\dot{\omega} = -\frac{2v \dot{\phi}}{\rho}$$

$$\dot{\rho} = v$$

$$\dot{\phi} = \omega$$

4 1st order ODEs to solve